

SHUFAN SHAHI

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EDUCATION

Islamic University of Technology B.Sc. in Computer Science and Engineering Coursework: Algorithms, Machine Learning, Deep Learning, Databases, and Networks.	2022–2026 CGPA: 3.55
Chuadanga Govt. College Higher Secondary Certificate (HSC)	2021 GPA: 5.00/5.00

EXPERIENCE

Student Researcher <i>Elite Research Lab</i> Support research projects through literature reviews, experiments, and analysis.	2026–Present
Intern <i>Battery Low Interactive Ltd.</i> Learned Flutter development, REST API integration, and applications of fine-tuned ML models.	2025
Researcher <i>IUT Computer Vision Lab</i> Researched multimodal AI methods to improve integration and mitigate modality imbalance.	2025–2026
Senior Executive <i>Project Aero</i> Contributed to drone R&D by applying AI-based object tracking for international UAV competitions.	2025–2026

PROJECTS

DocuMind - RAG Document Q&A Assistant <i>Python, LangChain, LlamaIndex, ChromaDB, Ollama</i> 🐙 GitHub	
- Built a local RAG system for documents with cited answers and hallucination guardrails. - Compared 4 chunking methods and 2 embedding models; added hybrid BM25 and reranking.	
Drone Vision - Aerial Tracking <i>PyTorch, YOLOv11X, ByteTrack</i> 🐙 GitHub	
Fine-tuned YOLOv11X on VisDrone and used ByteTrack for human counting and multi-object tracking.	
BD Traffic Monitoring System - Traffic Analytics <i>PyTorch, YOLO11L, Flask, Docker</i> 🐙 GitHub 📺 Live	
Fine-tuned YOLO on Bangladesh traffic data to detect local vehicle types.	
Biztrack - Business Analytics <i>Next.js, Node.js, PostgreSQL</i> 🐙 GitHub 📺 Demo	
Built AI modules for real-time finance tracking, risk alerts, customer segmentation, and forecasting.	
Postfolio - Professional Networking <i>Spring Boot, Next.js, Docker, RabbitMQ, Stripe</i> 🐙 GitHub 📺 Demo	
Integrates social networking, portfolio and CV creation, job search, and AI interview preparation.	
Redrick Routinson - Campus Automation <i>Python, React, Flask, Firebase</i> 🐙 GitHub 📺 Demo	
Built GA scheduling, QR exam attendance, class swapping, events, forums, and lost-and-found.	

RESEARCH EXPERIENCES

EthMuSAM: Mutual Scoring with SAM for Zero-Shot Anomaly Segmentation 2026

- Combined DINOv2 patch embeddings with SAM-based mutual scoring to localize zero-shot anomalies.

Prototype-Guided Adaptive Feature Learning for Multimodal Emotion Recognition in Conversation (MERC) 2026

- Used adaptive weighting and modality prototypes to improve multimodal emotion recognition.

Rethinking Visual Features in Graph-Based Multimodal Emotion Recognition in Conversations (MERC): From Low-Level Pixels to Semantic Representations 2026

- Used semantic visual representations to improve graph-based multimodal emotion recognition.

TECHNICAL SKILLS

Programming Languages: Python, C++, Java

Data and Machine Learning: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas

Databases: MySQL, Firebase, PostgreSQL

Web Development: React, Tailwind CSS, Next.js, Spring Boot

Mobile Development: React Native, Flutter

Visualization: Matplotlib, Seaborn

Development Tools: Git, Linux

ACHIEVEMENTS, AWARDS AND SCHOLARSHIPS

- **5th Place, VAND4 International Competition** — Zero-Shot Industrial Track 2026
- **First Runner-up, IUT Coderush Hackathon** — Rapid team application development 2025
- **7th Place, BAIC Datathon** — Data preprocessing, visualization, and modeling 2025

REFERENCE

Shahriar Ivan

Assistant Professor, Department of CSE, IUT

Specialization: Machine Learning, Deep Learning, Computer Vision

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